Polar Integration

Questions 1–47

Question 1. Find each of the following:

a) $\int (2\cos 3x)^2 \, dx$

b) $\int (2\sin 3x)^2 \, dx$

c) $\int (1+\cos x)^2 \, dx$

d) $\int (1-\sin x)^2 \, dx$

e) $\int \sin^2\left(\frac{x}{6}\right) \, dx$

f) $\int \left(2+\cos\frac{x}{2}\right)^2 \, dx$

Question 2. Find the exact area enclosed by each of the following curves:

a) $r = 4+3\cos\theta$ for $0 \le \theta \le 2\pi$

b) $r = 5+2\sin\theta$ for $0 \le \theta \le 2\pi$

c) $r = a(1-\cos\theta)$ for $0 \le \theta \le 2\pi$, for positive constant a

d) $r = a(1-\sin\theta)$ for $0 \le \theta \le 2\pi$, for positive constant a

Question 3. In each of the following cases, use polar integration to find the area enclosed by the curve given, where a is a positive constant:

a) $r = a\cos\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

b) $r = a\sin\theta$ for $0 \leq \theta \leq \pi$

c) $r = a$

Question 4. In each of the following cases, use polar integration to find the area bounded by the curve and the initial line, where a is a known positive constant:

a) $r = ae^{\theta}$ for $0 \leq \theta \leq \pi$

b) $r = a\theta$ for $0 \leq \theta \leq \pi$

Question 5. Sketch $r = 2\sin 3\theta$ for $0 \leq \theta \leq \frac{\pi}{3}$.

b) On your diagram, shade the region for which $r = 2\sin 3\theta$ and $\theta \leq \frac{\pi}{6}$.

c) Find the exact area of the region you have shaded in part (b).

Question 6. Show that the curve $r = a\sin^2\theta$ for $0 \leq \theta \leq \pi$ for positive constant a consists of a single closed loop through the pole.

a) Sketch the curve $r = a\sin^2\theta$ for $0 \leq \theta \leq \pi$, where a is a positive constant.

b) Find, in terms of a, the exact area of the region enclosed by the curve.

Question 7. A student is attempting to find the area enclosed by the curve C given by $r^2 = 6\cos 3\theta$. Their working is shown below:

$$A = \frac{1}{2}\int_{-\pi}^{\pi} r^2\, d\theta = \frac{1}{2}\int_{-\pi}^{\pi} 6\cos 3\theta\, d\theta = \frac{1}{2}[2\sin 3\theta]_{-\pi}^{\pi} = 0.$$

a) Sketch the curve C.
b) Explain what the student has done wrong.
c) Find the area enclosed by one petal of the curve.
d) Hence, state the total area enclosed by the curve $r^2 = 6\cos 3\theta$.

Question 8. Find the area enclosed by each of the following curves, giving your answer in exact form in terms of a:

a) $r^2 = a\sin 2\theta$
b) $r^2 = a\cos 2\theta$
c) $r^2 = a\sin 3\theta$
d) $r^2 = a\cos 3\theta$

Question 9. Find the area enclosed by each of the following curves, giving your answer in exact form:

a) $r = 2\cos 3\theta$

b) $r = 3\sin 2\theta$

c) $r = a\cos 4\theta$ for positive constant a

d) $r = a\sin 5\theta$ for positive constant a

Question 10. Find the area enclosed by the curve $r = \frac{1}{\sqrt{\theta}}$, the half-lines $\theta = \frac{\pi}{6}$ and $\theta = \frac{11\pi}{6}$, and the initial line.

Question 11. Find the area of the inside loop enclosed by each of the following curves, giving your answer in exact form:

a) $r = 1 + 2\cos\theta$ for $0 \leq \theta < 2\pi$

b) $r = 2 - 4\sin\theta$ for $0 \leq \theta < 2\pi$

Question 12. Show that the area enclosed by the curve $r = a(p + q\sin\theta)$ for $0 \leq \theta < 2\pi$, where $p \geq q$ and a is a positive constant, is given by $\frac{1}{2}a^2(2p^2 + q^2)\pi$ units^2.

Question 13. For each curve, find the ratio of the area bounded by the curve above the initial line to the area bounded below it:

a) $r = 2(1 + \cos\theta)$ for $0 \leq \theta < 2\pi$

b) $r = 2(1 + \sin\theta)$ for $0 \leq \theta < 2\pi$

Question 14. A hole for a pond has been dug with vertical sides and even depth 0.8 metres. The horizontal cross-section is modelled by the curve with equation

$$r = \frac{2}{5}(4 + \sin 4\theta), \qquad 0 \leq \theta \leq 2\pi,$$

where r is measured in metres.

a) Find the volume of water it will take to completely fill the pond, giving your answer in exact form.

b) State a modelling assumption that you have made.

Question 15. A large paddling pool, with depth 0.4 metres and horizontal cross-section modelled by the curve with equation

$$r = 5 + \cos 4\theta, \qquad 0 \leq \theta < 2\pi,$$

where r is measured in metres. The paddling pool is initially empty, then the water is turned on and pours into the paddling pool at a rate of 40 litres per minute.

a) Determine how long it will take to completely fill the paddling pool, giving your answer to the nearest minute.
b) State a limitation of the model.

Question 16. A curve is given by $r = ae^{k\theta}$ for $\theta \geq 0$, where a and k are positive constants. Find the area enclosed by the curve and the lines:

a) $\theta = 0$ and $\theta = \frac{\pi}{4}$
b) $\theta = 2\pi$ and $\theta = \frac{9\pi}{4}$
c) $\theta = 4\pi$ and $\theta = \frac{17\pi}{4}$
d) Hence show that the successive areas between the curve and the lines $\theta = 2n\pi$ and $\theta = 2n\pi + \frac{\pi}{4}$ increase in a geometric sequence. State the common ratio.

Question 17. A curve is given by $r = 1 + \sinh\theta$ for $0 \leq \theta \leq \pi$ and consists of a single closed loop through the pole. Find the area enclosed by the curve and the initial line, giving your answer to three significant figures.

Question 18. A curve is given by $r = \sqrt{3}\sin\theta$ for $0 \leq \theta \leq k$ for constant k.

a) State the smallest possible value of k for it to form a single loop.
b) Find the greatest value of r.
c) Hence sketch the curve.
d) Find the area enclosed by the curve.

Question 19. A curve C_1 is given by $r = \sqrt{16 \sin 2\theta}$ for $0 \leq \theta \leq k$ for constant k.

a) State the smallest possible value of k for it to form a single loop.

b) Find the area enclosed by C_1 for your value of k.

P is the point on the curve such that r is a maximum.

c) Find the polar coordinates (r, θ) of P.

The matrix $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ represents the transformation T. T maps C_1 onto a curve C_2.

d) Show that the point P after it has been mapped by transformation T has an argument that satisfies $\tan\theta = \frac{2}{4}$.

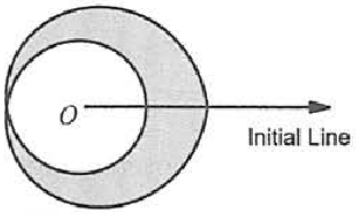
e) Find the area enclosed by C_2.

Question 20. A curve is given by polar equation $r = 2 + 4\cos\theta$ for $0 \leq \theta < 2\pi$.

a) Sketch the curve.

b) Show that the area of the region inside the curve, but not inside the loop, is $(4\pi + 12\sqrt{3})$ units^2.

Question 21. A curve is given by polar equation $r = 2 - \sin\left(\frac{\theta}{4}\right)$ for $0 \leq \theta \leq 4\pi$. Some regions enclosed by the curve as shown in the diagram have been shaded. Show that the area of the shaded region is $(-18 + 16\sqrt{2})$ units^2.



Question 22.

a) Given that $z = \cos\theta + i\sin\theta$, express $z^n + \frac{1}{z^n}$ in terms of trigonometric functions of $n\theta$.

b) By considering $\left(z + \frac{1}{z}\right)^4$, show that

$$\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}.$$

c) Sketch the curve with polar equation $r = a\cos^2\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

d) Find, in terms of a, the exact area of the region enclosed by the curve.

Question 23. A curve is given by $(x^2 + y^2)^3 = y^4$.

a) Find the polar equation of the curve.

b) Sketch the curve.

c) Find the area enclosed by the curve.

Question 24. A curve is given by $(x^2 + y^2)^2 = a^2(x^2 - y^2)$ for constant a.

a) Find the polar equation of the curve.

b) Sketch the curve.

c) Find, in terms of a, the exact area of two of the loops.

Question 25. A curve is given by $y^2 = a^2(x-y)^{4/3} - x^2$ for constant a.

a) Find the polar equation of the curve, giving your answer in the form $r^2 = f(\theta)$.

b) Show that $y = -x$ is a line of symmetry.

c) Sketch the curve.

d) Find, in terms of a, the exact area of two of the loops.

Question 26. A curve is given by

$$(x^2 + y^2)^2 + ax(3y^2 - x^2) = 0$$

for positive constant a.

a) Given that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, show that the polar equation of the curve is $r = a\cos 3\theta$.

b) Sketch the curve.

c) Find, in terms of a, the exact area of three of the loops.

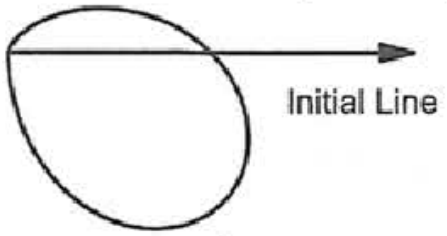
Question 27. A curve has polar equation $r = 3\cos\theta + \sin\theta$ for $0 \leq \theta \leq \pi$.

a) Find the area enclosed by the curve.

b) Find the Cartesian coordinates of the point that is furthest away from the pole.

c) Show that the curve is a circle whose radius and centre should be specified.

Question 28. The diagram shows the loop of the curve given by $r = 2(1 - \sin\theta)\cos\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Show that the area enclosed by the loop is $\frac{5\pi}{4}$ units^2.



Question 29.

a) Sketch the curve $r = a\sec\theta$ for positive constant a.

b) Use polar integration to find the area enclosed by $r = a\sec\theta$, the half-line $\theta = k$ for $0 < k < \frac{\pi}{2}$, and the initial line. Give your answer in terms of a and $\tan k$.

Question 30. Show that the area of the region enclosed by the curve given by

$$r = a\sec^2\left(\frac{\theta}{2}\right) \quad \text{for } 0 \leq \theta \leq \frac{\pi}{2},$$

for positive constant a, the initial line and the line $\theta = \frac{\pi}{4}$ is

$$a^2\left(\tan\frac{\pi}{8} + \frac{1}{3}\tan^3\frac{\pi}{8}\right) \text{ units}^2.$$

Question 31. A curve has polar equation

$$r = 2\sqrt{\cos\theta}\, e^{\frac{1}{3}\sin\theta}, \qquad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

a) Find the exact area enclosed by the curve.

b) Show that the greatest value of r on the curve is $2\sqrt{\frac{\sqrt{3}}{2}}e^{1/6}$.

Question 32. A polar curve is given by $r = \frac{12}{k} + k(2 + \sin\theta)$ for $0 \leq \theta \leq 2\pi$, for positive constant k.

a) Show that the area enclosed by the curve is $\left(\frac{9}{2}k^2 + 48 + \frac{144}{k^2}\right)\pi$ units^2.

b) Find the exact value of k such that the area is minimised.

Compound Areas

Question 33. Two curves C_1 and C_2 are defined by

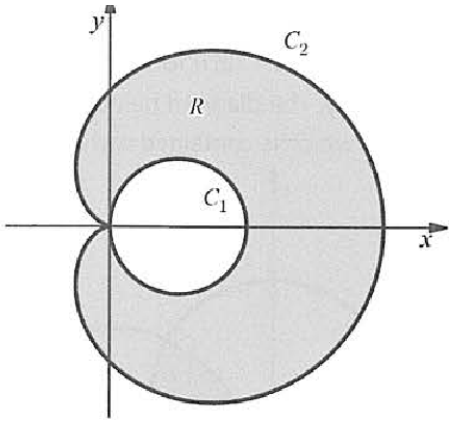
$$r_1 = \cos\theta \quad \text{for } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \qquad r_2 = a(1+\cos\theta) \quad \text{for } 0 \leq \theta \leq 2\pi,$$

respectively, where a is a positive constant.

a) Write down the polar coordinates (r, θ) of the points where curve C_1 crosses the initial line.

b) Explain why both C_1 and C_2 are symmetrical about the initial line.

c) Find the area R which lies inside C_2 but outside C_1, giving your answer in terms of a.



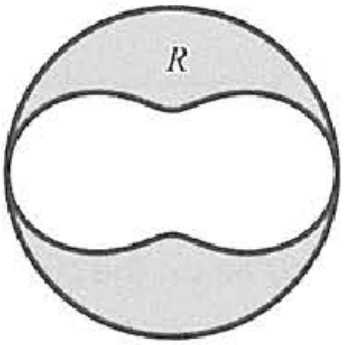
Question 34. A design of a logo can be formed by the polar curve with equation $r = 2.1 + a\cos(2\theta)$ for $0 \leq \theta \leq 2\pi$, where a is a constant, and a circle which meets the curve twice and is centred around the pole. The logo is designed to be 6 units across.

a) State the value of a.

b) Find the exact area of the shaded region of the logo.

The logo is to be painted on the wall of a building so that it is 12 m wide and requires two coats of paint, and a tin of paint covers 40 m^2.

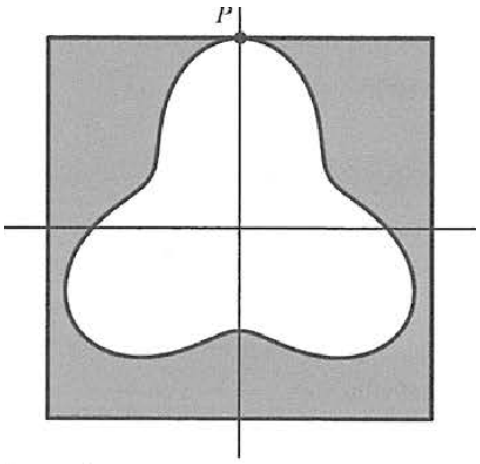
c) Determine how many tins of paint will be needed to paint the logo.



Question 35. An artist is designing a decorative panel by cutting a square piece of wood with side length 2.6 m. The curve to be removed is defined by polar equation $r = 1 + a\sin 3\theta$ for $0 \leq \theta \leq 2\pi$ and $-1 < a < 1$, and is positioned so it is symmetric and touches the top centre of the square, labelled as point P in the diagram below.

a) Show that $a = -0.3$.

b) Find the area of the remaining wood, giving your answer in m^2 to three significant figures.



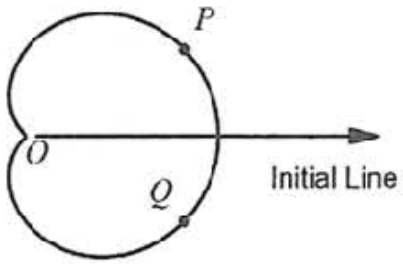
Question 36. The curve C shown in the diagram below is given by the polar equation $r = a + b\cos\theta$ for $-\pi \leq \theta \leq \pi$, where a and b are positive constants and $a > b$. Points P and Q lie on the curve C such that OPQ is an equilateral triangle which is symmetrical across the initial line.

a) Find the polar equation of the line passing through both P and Q in terms of a and b.

Given the point P occurs when $r = 2$ and $b = 1$:

b) State the exact value of a.

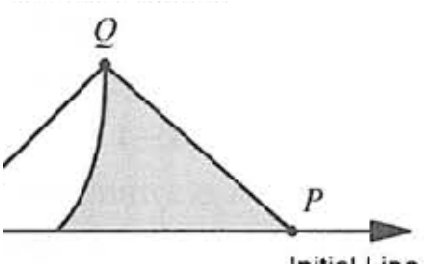
c) Show that the area of the equilateral triangle is approximately 30.9% of the area enclosed by the curve.



Question 37. The diagram shows curve C given by polar equation $r = 1 + \tan\theta$ for $0 \leq \theta \leq \frac{\pi}{4}$, the line segment OQ given by $\theta = \frac{\pi}{4}$ and the point P which has coordinates $(3, 0)$.

a) Find the length of line OQ.

b) Show that the exact area of the shaded region is $\frac{1}{2}(3\sqrt{2} - 1 - \ln 2)$.



Intersecting Polar Regions

Question 38. Two curves C_1 and C_2 are defined by

$$r_1 = 1 + \sin\theta \quad \text{for } 0 \leq \theta < 2\pi, \qquad r_2 = 3\sin\theta \quad \text{for } 0 \leq \theta < \pi,$$

respectively.

a) Sketch C_1 and C_2 on the same axes.

b) Find the point(s) of intersection(s) of the curves.

c) Explain whether or not the curves intersect at the pole.

Question 39. Two curves C_1 and C_2 are defined by

$$r_1 = 2(1 - \cos\theta) \quad \text{for } 0 \leq \theta < 2\pi, \qquad r_2 = 2\cos\theta \quad \text{for } -\frac{\pi}{2} \leq \theta < \frac{\pi}{2},$$

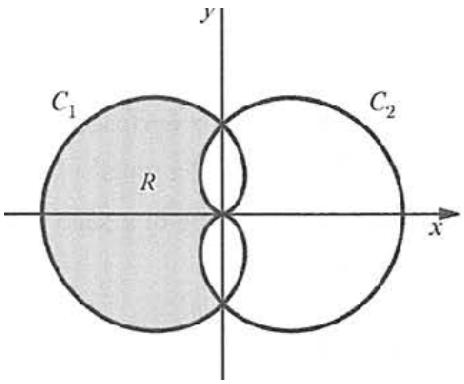
respectively.

a) Sketch C_1 and C_2 on the same axes.
b) Find the point(s) of intersection(s) of the curves.
c) Explain whether or not the curves intersect at the pole.

Question 40. Two curves C_1 and C_2 are defined by

$$r_1 = 1 - \cos\theta \quad \text{for } 0 \leq \theta < 2\pi, \qquad r_2 = 1 + \cos\theta \quad \text{for } 0 \leq \theta < 2\pi,$$

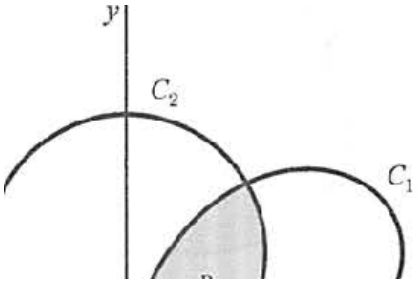
respectively. The diagram below shows the region R which lies inside C_1 but outside C_2. Determine the exact area of R.



Question 41. Two curves C_1 and C_2 are defined by

$$r_1 = \cos\theta\sqrt{2\sin\theta} \quad \text{for } 0 \leq \theta \leq \frac{\pi}{2}, \qquad r_2 = 3^{-1/4}\sin\theta \quad \text{for } 0 \leq \theta \leq \pi,$$

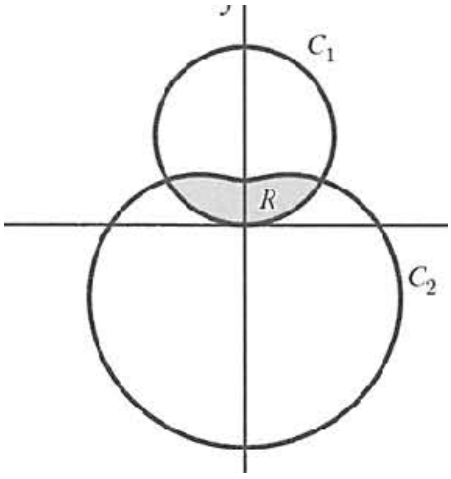
respectively. The diagram below shows the region R which is contained within both curves. Determine the exact area of R.



Question 42. Two curves C_1 and C_2 are defined by

$$r_1 = 4\sin\theta \quad \text{for } 0 \leq \theta < \pi, \qquad r_2 = 3 - 2\sin\theta \quad \text{for } 0 \leq \theta < 2\pi,$$

respectively. The diagram below shows the region R which is contained within both curves. Determine the exact area of R.

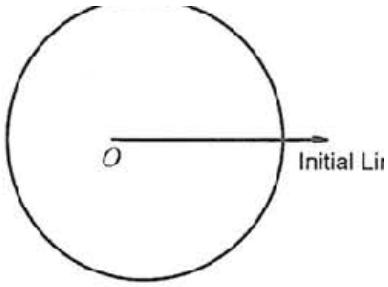


Question 43. The line L has polar equation $r = \frac{7\sqrt{3}}{4}\operatorname{cosec}\theta$ for $-\frac{\pi}{2} < \theta \leq \frac{\pi}{2}$ and the curve C has polar equation $r = 4 + \cos\theta$ for $-\pi \leq \theta \leq \pi$.

a) Verify that the two points of intersection occur when $\theta = \frac{2\pi}{3}$ and $\theta \approx 0.68817$.

b) On a copy of the diagram, add line L. The region R is the set of points which lie above the initial line such that $r < \frac{7\sqrt{3}}{4}\operatorname{cosec}\theta$ and $r < 4 + \cos\theta$.

c) Find the area of the region R, giving your answer to two significant figures.



Question 44. For each of the following pairs of curves, C_1 and C_2, find the area of the region that lies inside C_1 but outside C_2.

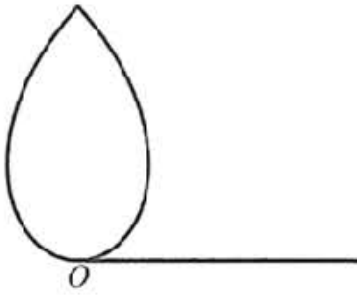
a) $r_1 = 3 + 2\sin\theta$ for $0 \leq \theta < 2\pi$ and $r_2 = 2$

b) $r_1 = 4 - 2\cos\theta$ for $0 \leq \theta \leq 2\pi$ and $r_2 = 6\cos\theta$ for $0 \leq \theta \leq \pi$

Question 45. The curve C is given by $r = 2\sinh\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. The diagram below shows the curve C.

a) On a copy of the diagram add the curve given by $r = 2$.

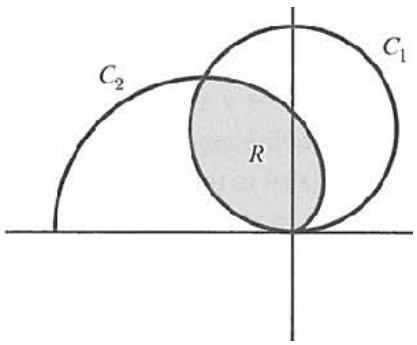
b) Show that the exact area of the region which lies inside both of the curves is $2\sqrt{2} + 2\pi - 6\ln(1+\sqrt{2})$ units^2.



Question 46. Two curves C_1 and C_2 are defined by

$$r_1 = 2\sin\theta \quad \text{for } 0 \leq \theta \leq \pi, \qquad r_2 = \frac{2\sqrt{3}}{3}(1-\cos\theta) \quad \text{for } 0 \leq \theta \leq \pi,$$

respectively. The diagram below shows the region R which is contained within both curves. Determine the exact area of R.



Question 47. Two curves C_1 and C_2 are defined by

$$r_1 = 2\cos\theta \quad \text{for } 0 \leq \theta \leq \frac{\pi}{2}, \qquad r_2 = 1 + \sin\theta \quad \text{for } 0 \leq \theta \leq \pi,$$

respectively. Show that an expression for the area of the region enclosed by C_1, C_2 and the initial line is $p\pi + q + r\phi$ units^2, where p, q and r are constants to be determined and $\sin\phi = \frac{3}{5}$.